Smart Garbage Bin

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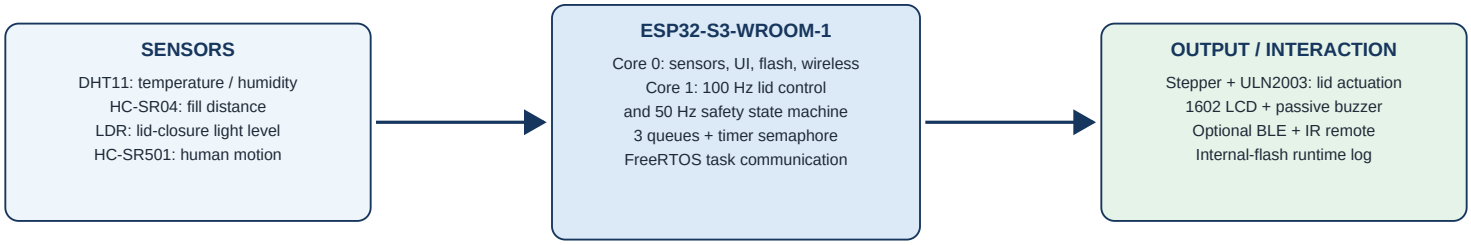
Description

The proposed **Smart Garbage Bin** is a dual-core ESP32-S3 system that automates lid operation while monitoring conditions inside the bin. A DHT11 measures temperature and humidity, an HC-SR04 estimates fill level, an LDR divider provides lid-closure feedback, and an HC-SR501 detects nearby human motion. The system opens the lid with a stepper motor driven through a ULN2003 module, reports live measurements on a 1602 I2C LCD, and uses a passive buzzer for full-bin and fault alerts. Internal flash retains measurements and event records for later analysis. Bluetooth Low Energy and an IR remote are planned only after the core sensor, actuator, display, logging, and safety functions are stable. The design emphasizes measurable real-time deadlines, task isolation, feedback-based closure verification, and a predictable safe state.

System Components and Data Flow



Category	Component / interface	Planned role and current status
Controller	ESP32S3 Dev Module; module marking ESP32-S3-WROOM-1	One board using both cores. Exact carrier manufacturer/model and revision remain unconfirmed.
Sensors	DHT11; HC-SR04; HC-SR501 PIR; LDR voltage divider	Temperature/humidity, fill distance, motion event, and lid-closure light level. LDR threshold will use calibration and hysteresis.
Actuator	Motion Pack stepper motor; ULN2003 driver; breadboard power module	Opens/closes the lid. Exact motor marking, rated voltage, mechanical linkage, and regulated external source remain TBD and will be verified before wiring.
Output / UI	1602 I2C LCD; passive buzzer	Standalone live display and audible status/fault patterns. Stepper-driven lid motion is treated as an actuator, not conservatively counted as a UI component.
Optional	On-chip BLE; acquired IR receiver + multi-key remote	Bidirectional data/commands over BLE and remote open/threshold commands over IR, only after the approved core functions pass V&V.;
Not baseline	SG90 servo	The project will not use the Motion Pack SG90 because the team does not expect it to provide adequate lid-opening torque.

Planned Runtime Architecture

Execution	Planned logic
Core 1 - real time	A hardware-timer ISR releases a binary semaphore for a **100 Hz LDR-feedback lid-control task**. A **50 Hz safety/state task** manages CLOSED, OPENING, HOLD-OPEN, CLOSING, FULL, and FAULT-SAFE states. During closing, new PIR motion requests reopening. Deadline and jitter evidence will be recorded under CPU load.
Core 0 - services	Low-rate DHT11 acquisition, HC-SR04 fill measurement while the lid is closed, LDR/PIR conditioning, LCD/buzzer updates, internal-flash logging, and optional BLE/IR handling. No Serial, LCD, allocation, delay, or sensor work occurs in an ISR.
Data/synchronization	**sensorDataQueue** carries filtered measurements; **controlCommandQueue** carries stepper direction/rate and open/close/setpoint commands; **systemEventQueue** carries motion, full-bin, timeout, and fault events. A timer-to-task binary semaphore wakes the control task.
Timing plan	Three ESP32 timing mechanisms are planned: 100 Hz control release (hardware timer ISR), 50 Hz safety supervision, and a low-rate environment/log schedule. The exact APIs counted as the three required ESP32 timers will be confirmed with the instructor before implementation.

Project Criteria Checklist

SUM26 criterion	How the proposal addresses it	Status
Reliable timing and I/O	Three planned timing mechanisms; at least one hardware-timer ISR. LDR uses hysteresis/temporal validation; IR/PIR events are filtered. Any added physical button will be debounced.	Plan
High performance	100 Hz lid-control task and 50 Hz safety/state task on Core 1; measured deadline/jitter results under synthetic CPU load.	Meets on plan
Novel hardware	DHT11, HC-SR04, and HC-SR501 are three new peripherals. The 1602 LCD provides I2C communication; whether its prior Lab 3 use affects this rubric item will be handled separately with the instructor.	**Pending**
Physical measurements	Temperature, humidity, fill distance/percentage, LDR light level, and PIR motion events.	Meets
User interface	The verbally approved standalone scope uses PIR-triggered human input, the LCD, passive-buzzer alerts, and visible actuated-lid feedback. The acquired IR multi-key remote and BLE client remain optional enhancements.	Meets on plan
Dual core	Core 1 owns real-time lid/safety work; Core 0 owns sensing, UI, logging, and optional wireless communication.	Meets on plan
Queues + semaphore	Three purpose-specific queues plus a timer-to-control binary semaphore.	Meets on plan
Closed-loop control	A 100 Hz feedback loop drives the stepper toward a calibrated, tunable closed-lid LDR threshold. Hysteresis and consecutive valid samples confirm closure; step/time limits produce a fault instead of continued motion.	Meets on plan
Logging + analysis	Internal-flash stream: time, temperature, humidity, fill percentage, LDR value, state, and faults. Derived metrics: moving-average fill level, opening count, timeout/fault count, and deadline-miss count.	Meets on plan
Fault-safe	Invalid/stale sensor data, lid-closure timeout, or task overrun produces FAULT-SAFE: cease further stepping, preserve a predictable lid state, sound an alert, show a fault code, and log the event.	Meets on plan
Supervised agent / V&V	Students approve every edit, perform wiring/upload/testing, and grant PASS only from hardware evidence. Each subsystem receives build, normal, boundary, and fault tests.	Meets on plan

Closed-Loop Lid Logic

At startup, the stepper moves slowly toward closed until the filtered LDR value reaches the calibrated closed-lid setpoint, establishing the home position. An open request then advances a bounded number of steps and holds the lid open. During closing, the 100 Hz task compares the filtered LDR feedback with the tunable setpoint and continues stepping only while the lid remains above the closure threshold. Hysteresis and consecutive valid samples prevent threshold chatter. Reaching the maximum allowed steps or timeout without closure stops further motion, logs the event, updates the LCD, and sounds the buzzer.

AI Collaboration Plan and Initial V&V

First subsystem: lid actuation and closure sensing. **Bounded request:** Goal - review the state machine and 100 Hz feedback task; Context - ESP32-S3-WROOM-1, Arduino IDE 2.3.7, Arduino-ESP32 3.3.10, Core 1 control, LDR feedback and ULN2003-driven stepper output; Constraints - no blocking in the control task, shared data only through queues, ISR only releases a semaphore, no wiring/pin/library changes without approval; Scope - plan and review only before any edit; Approval - Shawn and Franklin approve changes, wire, upload, and test; Acceptance/Evidence - the lid reaches the configured closed threshold without sustained hunting, records deadlines, and enters FAULT-SAFE on timeout, supported by build output, serial/flash logs, and hardware video or photographs.

Check	Procedure and acceptance criterion	Evidence / initial status
Build	Compile cleanly for ESP32S3 Dev Module with Arduino-ESP32 3.3.10.	IDE build output / NOT TESTED
Normal	PIR request opens, holds, and closes the lid; LDR confirms the calibrated closed threshold; no control deadline is missed.	Log + video / NOT TESTED
Boundary	Test calibrated light threshold margins and empty/full ultrasonic distances without rapid state oscillation.	Calibration log / NOT TESTED
Fault	Prevent closure or disconnect/stale a required sensor; system must stop repeated motion, alert the user, show a code, and log the event.	Fault log + video / NOT TESTED

References

[1] EE 474, *Final Project Guide SUM26*, course-provided PDF. [2] Espressif, ESP32-S3-WROOM-1/WROOM-1U Datasheet. [3] Aosong, DHT11 Datasheet. [4] HC-SR04 Datasheet. [5] FreeRTOS, Queues, Mutexes, and Semaphores. [6] Group 22 Lab 3 records: confirmed ESP32-S3-WROOM-1 module marking, Arduino IDE 2.3.7, Arduino-ESP32 3.3.10, and prior 1602 I2C LCD use.